

Communications and Networks (Spring 2022)

Problem Set 6

Issued: Mar. 23 Due: Jun. 5

Readings:

Forouzan - Data Communications and Networking (5th ed) Chapter 8, 12, 6.1.1, 6.1.3

Sklar - Digital Communications Fundamental and Applications (2nd ed) 11.1.4, 11.3.1-11.3.2

Proakis - Digital Communications (5th ed) 16.1-16.2

Bertsekas - Data Network (2nd ed) 4.2.2

1. Consider an FDMA system with $K = 2$ users, in an AWGN channel, where user 1 is assigned a bandwidth $W_1 = \alpha W$ and user 2 is assigned a bandwidth $W_2 = (1 - \alpha)W$, where $0 \leq \alpha \leq 1$. Let P_1 and P_2 be the average powers of the two users, while $N_0/2$ is the double-sided power spectrum density of the additive noise.

- a) Determine the capacities C_1 and C_2 of the two users and the total capacity $C = C_1 + C_2$ as a function of α .

Solution:

$$C = C_1 + C_2 = \alpha W \log_2 \left(1 + \frac{P_1}{\alpha W N_0} \right) + (1 - \alpha) W \log_2 \left(1 + \frac{P_2}{(1 - \alpha) W N_0} \right)$$

- b) Show that the maximum total capacity is achieved when $\alpha/(1 - \alpha) = P_1/P_2$.

Solution:

For simplicity, let $\lambda_i = P_i/WN_0$. Then, we obtain

$$\frac{dC}{d\alpha} = W \log_2 \left(\frac{1 + \frac{\lambda_1}{\alpha}}{1 + \frac{\lambda_2}{1-\alpha}} \right) + \frac{W}{\ln 2} \left(\left(1 + \frac{\lambda_2}{1-\alpha} \right)^{-1} - \left(1 + \frac{\lambda_1}{\alpha} \right)^{-1} \right). \quad (1)$$

Let $\frac{dC}{d\alpha} = 0$. Then, we can show that the maximum total capacity is achieved when $1 + \frac{\lambda_2}{1-\alpha} = 1 + \frac{\lambda_1}{\alpha}$. That is, $\alpha/(1 - \alpha) = P_1/P_2$.

2. (optional) In a pure ALOHA system, the channel bit rate is 2400 bits/s. Suppose that each terminal transmits a 100-bit frame every **minute**.

- a) Determine the maximum number of terminals that can use the channel without collision.

Solution:

For pure ALOHA, the vulnerable time is $2T_{fr}$. In this case, T_{fr} is given by

$$T_{fr} = \frac{100}{60 * 2400} \approx 6.94 \times 10^{-4} \text{s}. \quad (2)$$

Thus, the maximum number of terminals that can use the channel without collision is about $\frac{1}{2T_{fr}} \approx 720$.

- b) Repeat a) if slotted ALOHA is used.

Solution:

T_{fr} remains the same as a). However, the vulnerable time for slotted ALOHA is T_{fr} . So, the answer changes to $\frac{1}{T_{fr}} \approx 1440$.

3. In a multi-user slotted ALOHA system, the channel bit rate is 100 Mbps (1 M= 10^6). Assume that the arrival of packets follows a Poisson process. The size of a packet is 125 KB (1 Byte = 8 bits)

- a) Determine the total traffic arrival rate (in packet/s) that ensures the system achieving the maximum throughput.

Solution:

Assume the total arrival rate is λ_t .

Normalized total traffic:

$$G = \frac{b\lambda_t}{R} = \frac{8 \times 125 \times 10^3 \lambda_t}{100 \times 10^6} = \lambda_t \cdot 10^{-2}. \quad (3)$$

For slotted ALOHA, the normalized throughput is given by

$$\rho = Ge^{-G}, \quad (4)$$

which achieves its maximum 0.37 when $G = 1$.

Thus, the total traffic arrival rate (in packet/s) that ensures the system achieving the maximum throughput is $\lambda_t = 10^2$.

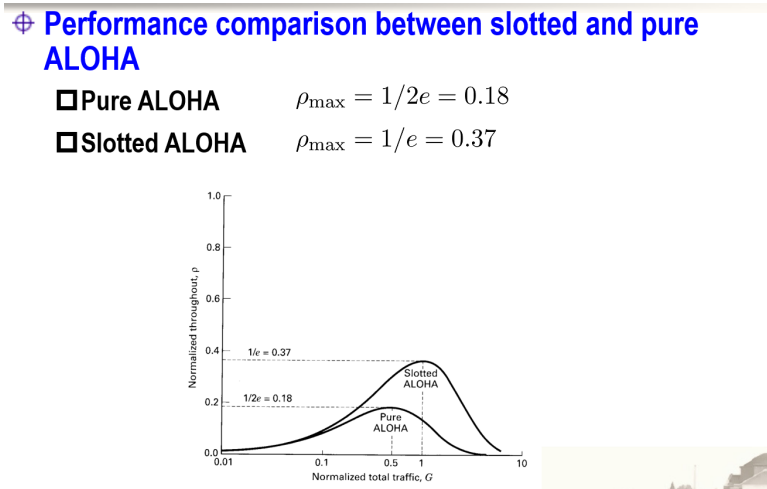
- b) Determine the total traffic arrival rate (in packet/s) with which the system's throughput is equal to half of the maximum throughput.

Solution: First we need to solve the G that satisfies

$$0.18 = Ge^{-G}. \quad (5)$$

Note that, there are two answers. The answers are $G \approx 0.225$ and $G \approx 2.72$. Thus, the total traffic rate is $\lambda_t = 0.225/10^{-2} = 22.5$ and $\lambda_t = 2.72/10^{-2} = 272$.

You can review the following figure in the slides, which can be used to solve both p3 b) and p4.



4. In a pure ALOHA network with $G = 1/2$, how is the normalized throughput affected in each of the following cases? (Increasing or decreasing?)

a) G is increased to 1.

Solution: Decreasing

b) G is decreased to $1/4$.

Solution: Decreasing

c) Repeat a) and b) if slotted ALOHA with $G = 1/2$ is used.

Solution: G is increased to 1: increasing

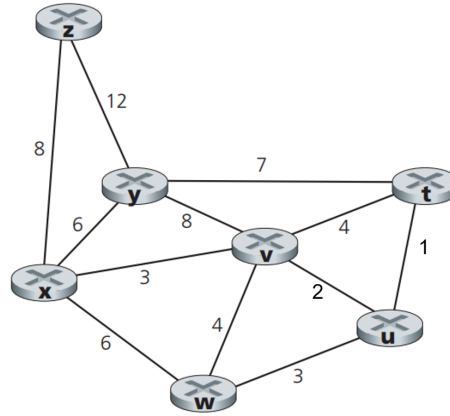
G is decreased to $1/4$: decreasing

5. List the strategies in CSMA/CA that are used to avoid collision. There is no acknowledgment mechanism in CSMA/CD, but we need this mechanism in CSMA/CA. Explain the reason briefly.

Solution: In CSMA/CA, collisions are avoided through the use of three strategies: the inter-frame space, contention window, and acknowledgments.

Acknowledgment is needed in CSMA/CA because the station does not have any means to check for the successful arrival of its data at the destination. On the other hand, the lack of collision in CSMA/CD is a kind of indication to the source that data have arrived.

6. Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes. Show how the algorithm works by computing the following table.



Step	start S	D(z), p(z)	D(y), p(y)	D(v), p(v)	D(w), p(w)	D(u), p(u)	D(t), p(t)
0	x						

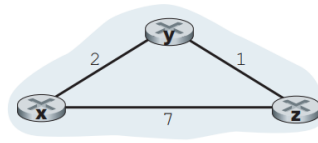
Solution:

Step	start S	D(z), p(z)	D(y), p(y)	D(v), p(v)	D(w), p(w)	D(u), p(u)	D(t), p(t)
0	x	8, x	6, x	3, x	6, x	∞	∞
1	x, v	8, x	6, x	3, x	6, x	5, v	7, v
2	x, v, u	8, x	6, x	3, x	6, x	5, v	6, u
3	x, v, u, y	8, x	6, x	3, x	6, x	5, v	6, u
4	x, v, u, y, w	8, x	6, x	3, x	6, x	5, v	6, u
5	x, v, u, y, w, t	8, x	6, x	3, x	6, x	5, v	6, u
6	x, v, u, y, w, t, z	8, x	6, x	3, x	6, x	5, v	6, u

7. Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table at node x (Hint: the initial distance table at node x is given below)

Solution: This is an example in Computer Networking (8th ed) by Kurose. The distance table at nodes x , y , and z are given by (on the next page)

Hope this example can help you understand the DV routing algorithm.



from	cost to		
	x	y	z
x	0	2	7
y	∞	∞	∞
z	∞	∞	∞

Time →

Node x table

from	cost to		
	x	y	z
x	0	2	7
y	∞	∞	∞
z	∞	∞	∞

Node y table

from	cost to		
	x	y	z
x	∞	∞	∞
y	2	0	1
z	∞	∞	∞

Node z table

from	cost to		
	x	y	z
x	∞	∞	∞
y	∞	∞	∞
z	7	1	0

..... Time →